

Rune Factory 4 Special / Rune Factory 5

Notes on Equipment Inheritance Mechanics

About This Document

This document is a set of research notes that organizes observations, tests, and measured results obtained during actual play of *Rune Factory 4 Special* (RF4SP) and *Rune Factory 5* (RF5), concerning:

- performance inheritance;
- arrangement inheritance;
- Auto-Arrange;
- Messhilite inheritance;
- Recursive Processing (nested references); and
- success-rate models.

The project originally began while the author was developing equipment for townspeople in RF5 with an emphasis on roleplay and consistency with the game world. During that process, the author noticed several behaviors that had not been sufficiently organized in existing information and began testing and recording them individually.

This document includes:

- observation results based on measured data;
- considerations that remain at the hypothesis stage;
- practical estimates of workload and success rates; and
- unresolved questions.

Some parts remain hypotheses and may be updated after further testing.

Readers are welcome to save this document locally and conduct additional testing or analysis as needed.

Nature of This Document

This document contains a mixture of:

- matters confirmed through measured observations;
- strong inferences based on observations; and
- untested hypotheses.

Accordingly, some of its organization is provisional and may be updated or changed in response to additional tests or new observations.

The document also reorganizes knowledge from actual-device testing and practical use while referring to testing and strategy information shared by Japanese and international communities.

In particular, many aspects of the following remain unresolved and may change after further testing:

- internal selection processing;
- recursive inheritance;
- weighting processes; and
- candidate priority.

About the Author

This document is based on the author's:

- testing on actual game platforms;
- observations during practical play; and

- organization of existing community knowledge.

At the time of writing, the author had played the *Rune Factory* series for approximately six months (approximately 500 hours in total). The document may therefore contain untested areas, mistaken observations, or interpretive differences.

The author is also unable to fully explain why the project developed into a document of this scale.

The initial purpose was simply to organize inheritance mechanics. Curiosity continued to expand the scope of testing, however, and the result became an extremely large collection of organized observations.

At present, the greatest unresolved question may not be RF inheritance mechanics themselves, but the mechanism governing the author's own curiosity.

Use of AI

AI was used in part to assist with organizing this document.

The purpose was not to delegate the work wholesale to AI. It was used primarily as a form of flood control: a way to organize and structure, to some degree, the information that continued to multiply through the author's own observations, tests, and reasoning.

The author therefore retained final responsibility for:

- testing decisions;
- decisions about what to adopt; and
- structural decisions.

Despite this flood-control effort, the final document still became extremely large. The author feels somewhat apologetic for the considerable load this also placed on the AI.

Reason for Publication

The author decided to publish this document because comparatively little publicly available information organizes inheritance mechanics in the *Rune Factory* series at this level of systematization, and because the document may contribute, even modestly, to organizing knowledge within the community.

To be candid, the author remains somewhat uncertain about whether a document of this nature should be made public.

Nevertheless, the author concluded that "Object X produced by digging this deeply might be better released into the world once than buried again," and published it despite some embarrassment.

Addendum

No particular restrictions are imposed on sharing, quoting, or circulating this document. However, because it contains many hypotheses, observations, and provisional arrangements, the author would appreciate it if modified redistributions identify the differences or modified passages wherever possible.

Because this is a long document containing many figures and tables, viewing it in print or on a large display is recommended where appropriate.

Table of Contents

1. Terminology

- 1-1. Performance Inheritance
- 1-2. Arrangement Inheritance
- 1-3. Auto-Arrange
- 1-4. Selection Processing
- 1-5. Becoming a Candidate

- 1-6. Performance-Source Contamination
- 1-7. Recursive Processing (Nested References)

2. Organization of Basic Mechanics

- 2-1. Overall Inheritance-Processing Flow
- 2-2. Separation of Performance Inheritance and Arrangement Inheritance
- 2-3. Components of the Candidate Count
- 2-4. Organized Differences between RF4SP and RF5
- 2-5. Organization of Boundary Values
- 2-6. Organization of Provisional Hypotheses

3. Detailed Analysis of Auto-Arrange

- 3-1. Trigger Conditions
- 3-2. Required Material Count and Slot Count
- 3-3. Random Selection Behavior
- 3-4. Inclusion of the First Material
- 3-5. Differences between RF4SP and RF5

4. Analysis of Performance-Source Contamination

- 4-1. Conditions Under Which Performance-Source Contamination Occurs
- 4-2. Number of Inheritances and Candidate Growth
- 4-3. RF4SP Observation Results
- 4-4. RF5 Observation Results
- 4-5. Differences between Shoes and Accessories
- 4-6. Order Independence and Changes in Candidate Order

5. Analysis of Recursive Processing

- 5-1. Definition of Nested Structures and Recursive Processing
- 5-2. References on the Performance-Inheritance Source Side
- 5-3. Tests on the Additional Arrangement-Material Side
- 5-4. Model of Internal Materials Becoming Candidates
- 5-5. The Recursion-Depth Question
- 5-6. Relationship with Internal Retention by Auto-Arrange
- 5-7. Effects on Split Saves and Separation of Work Stages

6. Analysis of Selection Processing

- 6-1. Processing after Candidate-Pool Generation
- 6-2. Candidate Count and Success Rate
- 6-3. Randomness in RF4SP
- 6-4. RF5 Behavior That Tends toward Fixed Results
- 6-5. Reselection When More Than Four Slots Are Involved
- 6-6. Boundary-Value Tests
- 6-7. The Question of How Order Is Determined

7. Mathematical and Generalized Models (Assumptions)

- 7-1. Generalized Candidate-Count Model
- 7-2. Approximate Success-Rate Model
- 7-3. Candidate Growth and Expected Values
- 7-4. Generalization of Boundary Values

- 7-5. Equal-Probability Selection Hypothesis
- 7-6. Mixing Different Types and the Permutation Problem
- 7-7. The Object X First-Position Problem
- 7-8. Weighted-Selection Hypothesis

8. Analysis of Messhilitte Inheritance

- 8-1. Overview of Messhilitte Inheritance
- 8-2. Conditions for Successful Inheritance
- 8-3. Relationship with Candidate Count and Same-Type Pool Manipulation
- 8-4. Relationship with Recursive Processing
- 8-5. Methods for Raising the Success Rate and Practical Strategies
- 8-6. Actual-Performance Contamination Problem
- 8-7. Practical Limits and Compromise Points

9. High-Difficulty Inheritance and Practical Use

- 9-1. The High-Difficulty Inheritance Problem
- 9-2. Split-Save Strategy
- 9-3. Candidate-Count Control
- 9-4. Practical Construction Strategies
- 9-5. Optimization of Townspeople's Equipment
- 9-6. Practical-Use Philosophy

10. Roleplay Equipment Research

- 10-1. Consistency with the Game World
- 10-2. Equipment for Individual Characters
- 10-3. Compatibility with Practical Performance
- 10-4. Design-Priority Inheritance
- 10-5. Coordination Examples
- 10-6. Romantic Builds and Practical Builds

11. Unresolved Questions and Future Testing Tasks

- 11-1. Selection Algorithm
- 11-2. Weighted-Selection Hypothesis
- 11-3. Internal-Priority Question
- 11-4. Recursion-Depth Limit
- 11-5. Differences between RF Titles
- 11-6. Optimization of High-Difficulty Inheritance Workflows

12. Addenda and Observation Materials

- 12-1. Gold Bracelet Test
- 12-2. Diamond Brooch Observation
- 12-3. Candidate-Count Boundary Test
- 12-4. Observation of Order Independence
- 12-5. AI Coordination Comparison
- 12-6. Additional Testing Notes

Appendices

Overview

Appendix A. Testing Environment and Observation Conditions

- A-1. Environment Used
- A-2. Position of Observation Counts in This Document
- A-3. Position of Screenshots in This Document
- A-4. List of Tests
- A-5. Items That May Be Required for Rigorous Testing

Appendix B. Observation Cases and Supplemental Materials

- Gold Bracelet Observation
- Diamond Brooch
- Conditions A/B/C
- Townspeople's Equipment Table
- Equipment List by Purpose

Appendix C. Candidate-Pool-Size Problems in the Rune Factory Series (Supplement)

Appendix A. Testing Environment and Observation Conditions

A-1. Environment Used

- RF4SP / RF5
- Nintendo Switch / Steam
- Save data used for testing
- Confirmation through Barrett / the smithy appraisal service
- Confirmation of blue text

A-2. Position of Observation Counts in This Document

The principal purposes of this document are to organize the structure of equipment-inheritance mechanics and to understand tendencies relevant to practical play.

Its position therefore differs from that of large-scale trials intended for academic statistical analysis or rigorous proof of random-number distributions.

In particular, many aspects of the following remain unresolved:

- internal selection algorithms;
- random-number seeds;
- candidate weights; and
- order-determination processing.

In theory, even 100,000 or more trials might not completely prove the internal implementation.

On the other hand, as observations of tendencies relevant to practical play, there were cases in which approximately 10 to 30 trials were sufficient to identify perceptible differences or behavioral tendencies in matters such as:

- lower success rates as the candidate count increases;
- order independence;
- tendencies toward Performance-Source Contamination; and
- the presence or absence of recursive references.

This document focuses on tendency observation through small numbers of trials and on structural organization. Rigorous statistical proof remains a task for future additional testing.

A-3. Position of Screenshots in This Document

The principal purposes of this document are to organize reproducibility conditions and analyze the structure of equipment-inheritance mechanics.

It therefore prioritizes clarification of the following over proving each individual test result with a single screenshot:

- reproduction procedures;
- equipment configurations;
- inheritance order;
- candidate conditions; and
- observed tendencies.

In particular, behaviors such as the following often cannot be fully explained by a single screenshot:

- changes in candidate order;
- Performance-Source Contamination;
- Auto-Arrange; and
- recursive references.

Accordingly, this document emphasizes the organization of reproducible test conditions and treats screenshots as the minimum necessary supporting material.

Future additional or external testing may organize observation logs, videos, or additional screenshots as required.

A-4. List of Tests

Test	Primary Subject	Connected Chapters
Gold Bracelet order-independence observation	Order independence in RF4SP	Chapters 4 and 6
RF5 recursive comparison	Recursive references	Chapters 5 and 6
Auto-Arrange boundary test	Candidate priority	Chapters 5 and 6
Messhilite inheritance	Candidate count and success rate	Chapters 6 and 8
Feather Boots test	Expansion of internal materials	Chapter 5
Performance-Source Contamination observation	Candidate re-registration	Chapter 4

A-5. Items That May Be Required for Rigorous Testing

This document primarily aims to organize the structure of equipment-inheritance mechanics and understand tendencies in practical use. It is not intended to provide rigorous statistical proof.

A-5-1. Statistical Methods That May Be Required for Rigorous Testing

Method	Main Purpose	Null Hypotheses	Example Relationship to This Document	Notes
Standard t-test (Student's t-test)	Detecting a difference in means	The difference is zero	Difference in success rates between RF4SP and RF5	The test statistic and standard-error formula differ depending on whether sample sizes are equal or unequal. Because this method assumes homogeneity of variance, Welch's t-test is often easier to use in practice.

Method	Main Purpose	Null Hypotheses	Example Relationship to This Document	Notes
Welch's t-test	Comparing means when variances differ	The difference is zero	Comparison of Conditions A/B/C	Because it does not assume homogeneity of variance, it is relatively compatible with heterogeneous game-testing data.
F-test	Comparing variances	The variances are equal	Difference in success rates between RF4SP and RF5; comparison of Conditions A/B/C	Can be used to check homogeneity of variance before a t-test. It can become unstable when the number of trials is insufficient.
Normality tests (general)	Checking data distributions	The data follow a normal distribution	Success-rate and candidate-count comparisons between RF4SP and RF5	Can be used to check assumptions for parametric tests such as t-tests and ANOVA. Game tests are easily affected by small sample sizes and skewed distributions.
Shapiro–Wilk test	Checking normality in small to medium datasets	The data follow a normal distribution	Comparison of Conditions A/B/C	Has relatively high power even for small samples and is often recommended; it is comparatively compatible with the scale of game testing.
Kolmogorov–Smirnov test	Checking agreement with a distribution	The data follow a specified distribution	Comparison of candidate-order distributions	Suited to large samples, but sensitive to differences in the tails of distributions.
Lilliefors test	Checking normality when the population mean and variance are unknown	The data follow a normal distribution	Comparison of success rates by candidate count	A modified K–S test that can be used when the population mean and variance are unknown.
Proportion test	Testing differences in success rates	There is no difference in success rates	Comparison of Messhilite success rates	Can be used to compare binary success/failure data.
Chi-square test	Testing equal-probability behavior	There is no distributional difference	Equal distribution of candidate order	Can be used to examine whether arrangement or selection order follows a uniform distribution.
Analysis of variance (ANOVA)	Comparing multiple conditions	All condition means are the same	Comparison among candidate counts N	Can be used for comparisons among multiple groups, such as differences by candidate count.
Nonparametric tests	Comparing non-normal distributions	There is no distributional difference	Comparison of order independence	A comparative method when a normal distribution is difficult to assume; a fallback option when trials are insufficient.
Least-squares method	Constructing an approximation model	There is no linear relationship	Approximation of candidate count versus success rate	May be used to model changes in success rate as the candidate count increases.
TOST (equivalence test)	Demonstrating the absence of a practically meaningful difference	The difference lies outside the acceptable range	Testing RF5 behavior that tends toward fixed results	Tests practical equivalence rather than merely whether a difference exists; compatible with analysis of fixed-result tendencies.
Non-inferiority test	Demonstrating that one condition is not worse	The difference exceeds the non-inferiority margin	Comparison of same-type pool manipulation	Can be used to determine not whether a condition improves, but whether it at least does not worsen the result.
Equal-probability test	Checking equal-probability selection	Selection probabilities are not equal	Analysis of arrangement and selection order	Can be used to examine whether selection candidates are weighted.

Note

This document primarily aims at structural organization and observation of practical tendencies, rather than rigorous proof based on large-scale statistical trials.

Many of the methods above are therefore organized as methods that might theoretically be required.

This table lists statistical methods that might theoretically be required if rigorous statistical proof were attempted for equipment-inheritance mechanics in RF4SP and RF5.

The principal purposes of this document are:

- structural organization;
- understanding observed tendencies; and
- analysis of practical use.

Full-scale statistical analysis using the methods above has not been performed.

In particular, rigorously proving the following statistically may require normality checks, checks for homogeneity of variance, and a sufficient number of trials:

- equal probability in candidate order;
- the significance of differences between RF4SP and RF5;
- rigorous examination of behavior that tends toward fixed results; and
- rejection of weighted selection.

On the other hand, there were cases in which observed tendencies or perceptible differences could be recognized with a comparatively small number of trials, including:

- lower success rates as the candidate count increases;
- RF5 behavior that tends toward fixed results when $N \leq 3$;
- instability around $N \geq 4$; and
- order independence in RF4SP.

This document therefore prioritizes observation-based structural organization and understanding of tendencies in practical use over rigorous statistical proof.

A-5-2. Statistical Quantities That May Be Required for Rigorous Testing

Statistical Quantity	Main Use	Example Relationship to This Document
MAE (mean absolute error)	Measuring discrepancies between theoretical and measured values	Candidate-count versus success-rate model
MAD (mean absolute deviation)	Measuring variation in data	Order independence in RF4SP
Mean	Mean success rate	Messhilité success rate
Median	Reducing the influence of outliers	Comparison of trial counts
Mode	Most frequently occurring result	RF5 behavior that tends toward fixed results
Variance	Variation in results	Order independence in RF4SP
Standard deviation	Measuring the range of variation	Comparison at candidate counts of four or more
Standard error	Estimation precision	Success-rate estimation
Correlation coefficient	Relationship analysis	Candidate count versus success rate
Confidence interval	Displaying an estimation range	Comparison of boundary values
p-value	Determining statistical significance	Comparison of differences between RF titles
Effect size	Practical importance	Effect of same-type pool manipulation

This document has not conducted rigorous statistical analysis using these quantities. Its principal purposes are structural organization, tendency observation, and practical-use analysis.

The document also uses a simplified expected-value model for the relationship between candidate count and success rate.

If a large amount of measured data is collected in the future, MAE (mean absolute error), among other quantities, may be used to evaluate the discrepancy between theoretical and observed values.

Appendix B. Observation Cases and Supplemental Materials

- Gold Bracelet observation
 - Diamond Brooch
 - Conditions A/B/C
 - Townspeople's equipment table
 - Equipment list by purpose
-

Appendix C. Candidate-Pool-Size Problems in the Rune Factory Series (Supplement)

Outside equipment inheritance, there appear to be cases elsewhere in the *Rune Factory* series in which the candidate pool grows as more items are shipped, including inventories at:

- general stores;
- flower shops;
- fish vendors; and
- material shops run by Raven in RF4SP or Sakuya in RF3.

For example, when attempting to obtain items such as Turnip Heaven or the ingredients for Gold Juice, shipping may increase the number of possible sale items and consequently appear to reduce the appearance rate of the target item.

This document has not tested these matters rigorously.

However, like Messhilite inheritance, Performance-Source Contamination, and candidate-compression problems, they may share a structural similarity from the perspective of the target candidate being diluted by growth in the candidate pool.

Accordingly, the operational principle of avoiding unnecessary growth in the candidate pool may have some significance across the *Rune Factory* series.

Ship items according to a plan.

Once an item has been shipped, it may become difficult to remove from the sale-candidate pool in that save data, or it may continue to remain a candidate.

At least in the author's environment, save data from playthroughs in which unnecessary crop shipping was restricted showed cases where Turnip Heaven and the ingredients for Gold Juice appeared in store inventories comparatively frequently.

This document has not, however, conducted rigorous testing of these observations.

Reading Guide

For Readers Seeking Practical Use

Chapters 7–10 are recommended: mathematical models, practical use, Messhilite, and roleplay.

For Readers Interested in Analysis of Inheritance Mechanics

Chapters 2–6 are recommended: basic mechanics, Auto-Arrange, candidate generation, Recursive Processing, and Selection Processing.

For Readers Interested in Mathematical and Success-Rate Models

Chapter 7 is recommended.

For Readers Interested in Unresolved Questions and Organized Hypotheses

Chapter 11 is recommended.

File Structure

Entry Documents

- 00_README_Inheritance_Research_EN
- 00_Summary_EN

Main Documents

- 01_Terminology_EN
- 02_Basic_Mechanics_EN
- 03_Auto_Arrange_EN
- 04_Performance_Source_Contamination_EN
- 05_Recursive_Processing_EN
- 06_Selection_Processing_EN
- 07_Mathematical_and_Generalized_Model_EN
- 08_Messhilite_Inheritance_EN
- 09_High_Difficulty_Inheritance_and_Practical_Use_EN
- 10_Roleplay_Equipment_Research_EN
- 11_Unresolved_Questions_EN
- 12_Observation_Cases_and_Supplement_EN

Research Appendices

- 99_AI_Coordination_Examples_EN
- 99_AI_Comparison_Log_EN
- 99_Equipment_List_by_Purpose_EN
- 99_Spreadsheet_Source_Data_CSV (folder)

Translation Authority Note

This file is a full English translation layer corresponding to the Japanese source document 00_README_. It does not replace the Japanese source as the primary authority. If a translation ambiguity is discovered, return to the Japanese source and preserve the original distinction among Observation, Hypothesis, and Unknown. Nothing in this translation should be read as a claim to have proven the game's internal implementation unless the Japanese source explicitly supports that claim.